

Kangyou Yu

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Research Interest

Human-Computer Interaction (HCI), AR/VR/MR, Adaptive User Interfaces, Multimodal Interaction, Input Techniques, Body Interaction, Exergame...

Education

- 2026.04-present **University of Melbourne**, Melbourne, Australia
Ph.D. in Engineering and IT
Advisor: [Adélaïde Genay](#) and [Jorge Goncalves](#)
- 2022.09-2024.12 **University of California, Santa Barbara (UCSB)**, Santa Barbara, USA
Ph.D. in Computer Science (transferred)
M.S. in Computer Science
Advisor: [Tobias Höllerer](#)
- 2018.09-2022.07 **Xi'an Jiaotong-Liverpool University (XJTLU)**, Suzhou, China
University of Liverpool (UoL), Liverpool, United Kingdom
B.Sc. in Information and Computing Science (**With Honors**)
Advisor: [Hai-Ning Liang](#)

Full Publications

CONFERENCE PAPERS

- [C.8] **Kangyou Yu**, Yunhao Luo, Radha Kumaran, Shane Dirksen, Misha Sra, Tobias Höllerer. *"The Cost of Virtuality Switching: Searching for Physical and Virtual Targets in Optical-See-Through Augmented Reality"* **IEEE ISMAR 2025**
- [C.7] Chengyuan Xu, Radha Kumaran, Noah Stier, **Kangyou Yu**, Tobias Höllerer. *"Multimodal 3D Fusion and In-Situ Learning for Spatially Aware AI"* **IEEE ISMAR 2024**
- [C.6] Xuetong Wang, Ziyang Wang, Mingmin Zhang, **Kangyou Yu**, Pan Hui, Mingming Fan. *"Avatar Appearance and Behavior of Potential Harassers Affect Users' Perceptions and Response Strategies in Social Virtual Reality (VR): A Mixed-Methods Study"* **CSCW 2024**

- [C.5] Abby Wysopal, Vivian Ross, Joyce Passananti, **Kangyou Yu**, Brandon Huynh, Tobias Höllerer. *"Level-of-Detail AR: Dynamically Adjusting Augmented Reality Level of Detail Based on Visual Angle."* **IEEE VR 2023**
- [C.4] Anil Ufuk Batmaz, **Kangyou Yu**, Hai-Ning Liang, Wolfgang Stuerzlinger. *"Improving Effective Throughput Performance using Auditory Feedback in Virtual Reality."* **ACM SUI 2022**
- [C.3] Wenge Xu, Xuanru Meng, **Kangyou Yu**, Hai-Ning Liang, Sayan Sarcar. *"Evaluation of Text Selection Techniques in Virtual Reality Head-Mounted Displays."* **IEEE ISMAR 2022**
- [C.2] Wenge Xu, Hai-Ning Liang, **Kangyou Yu**, Nilufar Baghaei. *"Effect of Gameplay Uncertainty, Display Type, and Age on Virtual Reality Exergames."* **ACM CHI 2021**
- [C.1] Wenge Xu, Hai-Ning Liang, Yuzheng Chen, Xiang Li, **Kangyou Yu**. *"Exploring Visual Techniques for Boundary Awareness During Interaction in Augmented Reality Head-Mounted Displays."* **IEEE VR 2020, Best Paper Nomination (5%)**

JOURNAL ARTICLES

- [J.6] Tom Bullock, Emily Machniak, Joyce Passananti, **Kangyou Yu**, Radha Kumaran, You-Jin Kim, Jordan Garrett, Madhav Viswesvaran, Julia Ram, Tobias Höllerer, Barry Giesbrecht. *"Electrophysiological evidence for differential semantic processing of words and objects presented in augmented reality."*
Computers in Human Behavior Reports
- [J.5] Wenge Xu, **Kangyou Yu**, Xuanru Meng, Diego Vilela Monteiro, Dominic Kao, Hai-Ning Liang. *"Exploring the effect of the Group Size and Feedback of non-player character spectators in virtual reality exergames."*
Frontiers in Psychology (Human-Media Interaction section)
- [J.4] **Kangyou Yu**, Shaoyue Wen, Wenge Xu, Maurizio Caon, Nilufar Baghaei, Hai-Ning Liang. *"Cheer for Me: Effect of Non-Player Character Audience Feedback on Older Adult Users of Virtual Reality Exergames."* **Springer Virtual Reality (VIRE) 2023**
- [J.3] Wenge Xu, Hai-Ning Liang, **Kangyou Yu**, Shaoyue Wen, Nilufar Baghaeil, Huawei Tu. *"Acceptance of Virtual Reality Exergame Among Chinese Older Adults."*
International Journal of Human-Computer Interaction 2021
- [J.2] Wenge Xu, Hai-Ning Liang, Nilufar Baghaei, Xiaoyue Ma, **Kangyou Yu** Xuanru Meng, Shaoyue Wen. *"Effects of an Immersive Virtual Reality Exergame on University Students' Anxiety, Depression, and Perceived Stress: Pilot Feasibility and Usability Study."* **JMIR Serious Game 2021**
- [J.1] Wenge Xu, Hai-Ning Liang, Qiuyu He, Xiang Li, **Kangyou Yu**, Yuzheng Chen. *"Results and Guidelines From a Repeated-Measures Design Experiment Comparing Standing and Seated Full-Body Gesture-Based Immersive Virtual Reality Exergames: Within-Subjects Evaluation."* **JMIR Serious Game 2020**

Research Experience

- 04/2026-present **The University of Melbourne (UniMelb), Melbourne, AU**
Graduate Researcher
HCI Group, Advisor: [Adelaide Genay](#) and [Jorge Goncalves](#)
Working on the topic of HCI and XR.
- 10/2022-10/2025 **UC-Santa Barbara (UCSB), Santa Barbara, CA**
Graduate Research Assistant
FourEyesLab, Advisor: [Tobias Höllerer](#)
Worked on the topic of HCI and mobile MR, focus on the input modalities and input techniques in mobile Mixed Reality.
- Worked on a research that is combining the AR and EEG staffs to investigate if we can detect significant N400 human brain signal in classifying word-word pairs and word-object pairs while wearing AR glasses. [J.6]
 - Leading a research that investigates the difference between the selection of physical objects and virtual objects in mixed reality. [C.8]
 - I worked on spatially aware AI for the mixed reality. [C.7]
 - I worked on designing dynamically adjusting augmented reality level of details in mixed reality. [C.5]
- 06/2022-10/2022 **The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China**
Undergraduate Research Intern
APEX Lab, Advisor: [Mingming Fan](#)
Worked on a project which examines the moderating effect of avatar appearance on user perception of sexual harassment and uncovering the underlying reasons behind response strategies. ([C.6])
- 04/2021-10/2022 **Simon Fraser University (SFU), Vancouver, Canada**
Undergraduate Research Intern
VVISE Lab, Advisor: [Wolfgang Stuerzlinger](#) and [Anil Ufuk Batmaz](#)
- I worked as an undergraduate research intern
 - I co-designed the experiment and collected data by conducting two user studies and analyzed the data
 - The project published at ACM SUI 2022 [C.4]
- 04/2019-07/2022 **Xi'an Jiaotong-Liverpool University (XJTLU), Suzhou, China**
Undergraduate Research Assistant
X-CHI Lab, Advisor: [Hai-Ning Liang](#) and [Wenge Xu](#)
- I worked as an undergraduate research assistant

- I worked on evaluating the text selection techniques in VR. [C.3]
- I worked on investigating the effect of different factors on different ages groups of users in VR exergames. [J.1][J.2][J.3][J.4][J.5][C.2]
- I worked on exploring visual techniques for boundary awareness in AR HMDs. [C.1]

Awards and Honors

2026 Melbourne Research Scholarship (AU\$39,500 per year)
 2023 SIGCHI Gray Marsden Travel Award (\$2000)
 2022 Academic Excellence Fellowship from UCSB (\$2000)
 2021 ACM UIST 2021 Registration Scholarship
 2020 Research Assistant Fellowship at XJTLU (\$1,400)
 2020 IEEE VR & 3DUI 2020 Best Conference Paper Nominee (5%)
 2019 Summer Undergraduate Research Fellowship (SURF) at XJTLU

Teaching Experience

04/2024-06/2024 **Teaching Assistant @ UCSB-CMPSC 16 Problem Solving I**
 04/2023-06/2023 **Teaching Assistant @ UCSB-CMPSC 9 Intermediate Python**

Professional Services

Reviewer: ACM UIST 2026, ACM CHI 2026, ACM CHI 2025, IEEE ISMAR 2025, ACM CHI 2024, ACM CHI Play 2024, ACM VRST 2024, ACM CHI 2023, IJHCI 2023

Student Volunteer: IEEE ISMAR 2024, IEEE ISMAR 2023, ACM CHI 2023, IEEE VR 2023, ACM SUI 2022, IEEE ISMAR 2021, ACM IUI 2021, IEEE AIVR 2020